

Lecture 16: The Physics of Control (Delayed Neutrons and the Inhour Equation)

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Reading Assignment

Lamarsh & Baratta (4th Edition):

- **Section 7.2** Reactor Kinetics (Prompt/Delayed Neutrons, Inhour Equation, Prompt Critical).

Recap: The Problem with Prompt Neutrons

In Lecture 15, we saw that if a reactor relied solely on prompt neutrons, the time scale of the reaction (the Reactor Period T) would be on the order of milliseconds.

$$T \approx \frac{l}{\rho} \quad \text{where} \quad \rho = \frac{k-1}{k}$$

If $l = 10^{-4}$ s and $\rho = 0.001$, the power rises by a factor of 22,000 in one second. This is uncontrollable. Today, we mathematically prove how **Delayed Neutrons** save us.

1 The Point Kinetics Equations

To account for delayed neutrons, we must modify our neutron balance equation. We split the neutron source into two terms:

1. **Prompt Neutrons:** $(1 - \beta)$ fraction of fission neutrons appear instantly.
2. **Delayed Neutrons:** β fraction of fission neutrons appear later, coming from the decay of Precursors (C).

We now need two coupled differential equations: one for the Neutrons (n) and one for the Precursors (C).

1.1 1. The Neutron Equation

$$\frac{dn}{dt} = \underbrace{\frac{k(1-\beta)n}{l}}_{\text{Prompt Generation}} + \underbrace{\lambda C}_{\text{Delayed Source}} - \underbrace{\frac{n}{l}}_{\text{Loss}}$$

Here, λ is the decay constant of the precursor (e.g., ^{87}Br).

1.2 2. The Precursor Equation

The precursors are created by fission and lost by radioactive decay.

$$\frac{dC}{dt} = \underbrace{\frac{k\beta n}{l}}_{\text{Production}} - \underbrace{\lambda C}_{\text{Decay}}$$

2 The Solution: The Inhour Equation

We look for solutions of the form $n(t) = n_0 e^{t/T}$ (exponential growth with period T). Substituting this ansatz into the coupled system involves some algebra (omitted here for brevity, see Lamarsh Ch. 7), but the result is the fundamental equation of reactor control:

$$\rho = \frac{l}{Tk_{eff}} + \frac{\beta_{eff}}{1 + \lambda T} \quad (1)$$

The Inhour Equation (Simplified One-Group Form)

Historical Note: You might wonder who "Dr. Inhour" was. There is no such person! The term "Inhour" is actually a portmanteau of **In**verse **H**our. It was an early unit of reactivity defined by Enrico Fermi's team at Chicago Pile-1 as the amount of positive reactivity that would result in a stable reactor period of exactly one hour.

This equation relates the **Reactivity** (ρ , what the operator changes with control rods) to the **Reactor Period** (T , how fast the power changes). Recall the reactor period T is the time required for power to change by a factor of e .

Note: In a real reactor, there are 6 distinct groups of precursors with different decay constants λ_i . The full equation is a sum:

$$\rho = \frac{l}{T} + \sum_{i=1}^6 \frac{\beta_i}{1 + \lambda_i T}$$

3 Analyzing the Result

Let's analyze the One-Group equation to understand the three distinct regimes of reactor operation.

3.1 Regime 1: Sub-Prompt Critical (Normal Operation)

Suppose we insert a very small amount of reactivity: $\rho \ll \beta$. Assume $\rho = 0.001$ (0.1%). For U-235, $\beta \approx 0.0065$ (0.65%). Since l is tiny (10^{-4}), the first term l/T is negligible for large T . The equation simplifies to:

$$\rho \approx \frac{\beta}{1 + \lambda T}$$

Solving for Period T :

$$T \approx \frac{\beta - \rho}{\lambda \rho}$$

Using $\lambda \approx 0.1 \text{ s}^{-1}$ (average precursor decay constant):

$$T \approx \frac{0.0065 - 0.001}{(0.1)(0.001)} = \frac{0.0055}{0.0001} = \mathbf{55} \text{ seconds}$$

Result: Instead of 0.1 seconds (Prompt), the period is 55 seconds. This is slow! A human operator can easily watch a gauge and adjust a rod in 55 seconds.

3.2 Regime 2: The Threshold (Prompt Critical)

What happens if we add enough reactivity such that $\rho = \beta$?

$$\beta \approx \frac{l}{T} + \frac{\beta}{1 + \lambda T}$$

The math blows up. T must become very small for this equality to hold. Physically, this means the reactor is critical on prompt neutrons *alone*. The delayed neutrons are no longer needed to sustain the chain reaction.

3.3 Regime 3: Super-Prompt Critical (The Danger Zone)

If $\rho > \beta$, the reactor is supercritical on prompt neutrons.

$$\rho \approx \frac{l}{T}$$

We return to the "Bomb" kinetics of Lecture 15. The period drops to milliseconds with only a tiny increase in the reactivity.

$$T \approx \frac{10^{-4}}{0.001} = 0.1 \text{ s}$$

4 Units of Reactivity: The Dollar

Because the fraction β is the most important safety limit in a reactor, we use it as a unit of measurement.

- **1 Dollar (\$1.00)** = A reactivity of β .
- **1 Cent (1¢)** = $0.01 \times \beta$.

For U-235 ($\beta = 0.0065$):

- \$1.00 = $0.0065 \Delta k/k$
- \$0.50 = $0.00325 \Delta k/k$ (Safe)
- \$1.10 = $0.00715 \Delta k/k$ (Prompt Critical - Dangerous)

"The reactor is rising on a 50-cent period" is standard control room terminology.

4.1 Case Study: The Plutonium Problem

To understand why we use Dollars instead of just reporting absolute $\Delta k/k$, imagine two identical operators working on two different reactors. Operator A is running a standard U-235 reactor. Operator B is running an experimental Pu-239 Fast Reactor.

Both operators make the exact same minor error: they accidentally withdraw a rod that inserts an absolute reactivity of $\Delta k/k = 0.004$ (just 0.4%).

- **Operator A (U-235):** Since $\beta = 0.0065$, the reactivity is $\rho = 0.004/0.0065 = \$0.61$. Looking at the Inhour equation, this puts the reactor on a stable, easily manageable period of about 10 seconds. Operator A calmly inserts the rod back in.
- **Operator B (Pu-239):** Since $\beta = 0.0021$, the exact same rod movement yields $\rho = 0.004/0.0021 = \$1.90$. Operator B just went prompt critical, and the core melts in milliseconds.

The Lesson: Absolute reactivity ($\Delta k/k$) doesn't tell you how dangerous a reactor state is unless you know the fuel's specific β . 50 cents is identically safe in both reactors, even though the physical rod movements required to get there are vastly different.

5 Physical Insight: Why does it work?

Think of the delayed neutrons as a "storage buffer."

- When you increase reactivity, the prompt neutrons rise instantly.
- But the delayed neutrons (0.65% of the population) are "stuck" in the precursors (Bromine/Iodine atoms). They haven't been born yet.
- The total population cannot rise to the new equilibrium level until those precursors decay and release their neutrons.
- This "drag" holds the reactor back, forcing it to wait for the precursors (approx. 10 to 80 seconds).

6 Case Study: TRIGA Reactors and the Ultimate "Save"

If \$1.00 is the ultimate danger limit, how do university research reactors (like the TRIGA design) routinely "pulse" their reactors to \$2.00 or even \$3.00 on purpose?

In a TRIGA reactor, the operator fires a pneumatic cylinder that ejects a control rod in a fraction of a second, intentionally making the reactor prompt critical. The power skyrockets on a millisecond period, peaking at upwards of 22,000 MW, producing a brilliant Cherenkov blue flash.

Why doesn't it melt? In Lecture 15, we discussed how U-238 Doppler broadening is a "first line of defense," but that it's often not quite big enough to stop a massive prompt excursion before the fuel melts; the reactor relies on the water (moderator) heating up to fully shut down, which takes *seconds* due to heat conduction.

The TRIGA completely bypasses this conduction delay by using **Uranium Zirconium-Hydride (U-ZrH)** fuel. The hydrogen (the moderator) is atomically bound within the fuel matrix itself, acting as a quantum harmonic oscillator within the crystal lattice.

When the uranium fissions, it heats the bound hydrogen *instantly*. As the hydrogen lattice heats up, the atoms vibrate wildly. When thermal neutrons bounce off these hot hydrogen atoms, they gain a discrete quantum of energy (upscattering), instantly shifting the thermal neutron energy spectrum "warmer."

Because the U-235 fission cross-section strictly follows a $1/v$ dependence in this thermal region, these "warmed" neutrons suddenly have a drastically lower probability of causing fission. This spectrum hardening alone provides a massive, instantaneous negative reactivity insertion.

Furthermore, to allow modern TRIGAs to run for decades without refueling, the fuel is deliberately loaded with extra uranium and alloyed with a burnable poison called **Erbium-167**. Erbium has a massive absorption resonance around 0.46 eV. As the hot hydrogen pushes the neutrons to higher energies to escape U-235, they are instantly swallowed by the Erbium "wall."

Together, the $1/v$ drop-off and the Erbium resonance pull the core back well below β , safely shutting the pulse down in less than 30 milliseconds—long before the heat ever reaches the water coolant.

The Aftermath: Post-Pulse Physics and Timescales

A common student question is: *What happens after the blue flash?* If the temperature feedback is what shut the reactor down, won't it just go prompt critical again as soon as the fuel cools off in the water?

The answer comes down to a race between mechanical systems and thermodynamics. Here is the exact timeline of a TRIGA pulse:

- **t = 0 seconds:** The pneumatic cylinder fires, ejecting the transient control rod. ρ jumps to β .
- **t = 0.015 seconds (15 ms):** Reactor power peaks at $\approx 20,000$ MW. The fuel temperature spikes to $\approx 400^\circ\text{C}$.
- **t = 0.030 seconds (30 ms):** The Erbium-167 and U-ZrH temperature feedback crushes the chain reaction. Power drops by several orders of magnitude.

At this exact moment, the control rod is **still physically out of the core**. The reactor is being held subcritical entirely by its own heat. It settles into a "tail" of low, steady power driven by the decay of delayed neutron precursors, perfectly balanced by the hot fuel.

- **t = 1 to 2 seconds:** The pneumatic pressure in the transient rod cylinder is automatically vented. Gravity pulls the heavy control rod back down into the core, driving the reactivity deeply negative.
- **t = 10 to 20 seconds:** The thermal time constant of the U-ZrH fuel kicks in. Heat finally conducts across the cladding and into the pool water. You will see bubbles rise to the surface of the pool as the water immediately surrounding the elements boils.

The Engineering Takeaway: The shutdown physics (the negative temperature coefficient) operates in **milliseconds**, safely killing the excursion. However, the heat transfer to the coolant takes **seconds**. This massive disparity in timescales gives the mechanical safety systems (the gravity-drop of the control rod) plenty of time to re-insert the physical poison before the core can cool down and accidentally restart the chain reaction.

Why Build a TRIGA? Applications of Pulsing

While avoiding a meltdown is a great safety feature, the ability to pulse a reactor to 22,000 MW on purpose has incredible scientific and economic value.

1. Simulating Reactor Accidents (Transient Testing) To test how new commercial nuclear fuels or reactor components will survive a Reactivity Initiated Accident (RIA), engineers must subject them to a massive, instantaneous spike in neutron flux. Pulsing a TRIGA reactor

allows engineers to recreate the extreme conditions of a prompt critical excursion inside a safe, controlled test chamber.

2. High-Flux Access without High-Flux Costs A steady-state 1,000 MW commercial reactor requires massive cooling towers, gigantic pumps, and thick pressure vessels. A TRIGA can achieve the same neutron flux ($\approx 10^{16}$ n/cm²-s) for a few milliseconds while maintaining a time-averaged power of just a few kilowatts. This provides researchers with brief bursts of world-class neutron flux using nothing more than a natural-convection swimming pool for cooling.

3. Neutron Activation Analysis (NAA) Researchers can place a sample of unknown material (e.g., a geological core, forensic evidence, or an archaeological artifact) into the core and pulse the reactor. The intense burst of neutrons is absorbed by the sample, transmuting trace elements into unstable radioactive isotopes. By analyzing the specific spectrum of delayed gamma rays emitted as the sample decays, scientists can identify the exact elemental composition down to the parts-per-billion level without chemically destroying the sample.

4. Neutron Radiography Neutron radiography is similar to an X-ray, but it uses a beam of neutrons. While X-rays bounce off dense heavy metals and pass right through organics, neutrons easily penetrate heavy metals (like lead or steel) but are heavily scattered by light elements (like the hydrogen found in water, plastics, or oil). TRIGA neutron beams are used to take internal "pictures" of aircraft engine turbine blades to check for microscopic water or oil blockages inside solid titanium casings.

References for Further Reading:

- **General Atomics (GA):** *TRIGA Nuclear Reactors*. (GA has installed over 60 TRIGA reactors across 24 countries, primarily for universities and medical centers).
- **IAEA (2016):** *History, Development, and Future of TRIGA Research Reactors* (Technical Reports Series No. 482).

7 Summary

1. **Point Kinetics:** We couple the neutron population to the precursor concentration.
2. **The Inhour Equation:** Relates Reactivity (ρ) to Period (T).
3. **Safety Limit:** We must keep $\rho < \beta$ (less than \$1.00).
4. **Normal Operation:** As long as $\rho < \beta$, the reactor period is determined by the slow decay of precursors (λ), making the reactor controllable by humans.